

Kyle Wong

Recent Master's graduate from Sidney Sussex College, University of Cambridge

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GitHub: <https://github.com/kyleyhw>

Website: https://kyleyhw.github.io/personal_website/

EDUCATION

Sidney Sussex College, University of Cambridge , Cambridge, England, United Kingdom MAST/Part III in Astrophysics Graduated with Good Honours equivalent	2024 – 2025
Victoria College, University of Toronto , Toronto, Ontario, Canada HBSc in Physics & Mathematics Graduated with High Distinction	2020 – 2024
German Swiss International School , Hong Kong SAR, China International Baccalaureate (IB) Diploma in 6 subjects	2010 – 2020

RESEARCH EXPERIENCE

Master of Advanced Study Research Project <i>Postgraduate Student</i>	Cambridge, England, United Kingdom October 2024 – September 2025
Implemented and statistically validated variable cosmological initial conditions in the 21cmSPACE simulation package, enabling efficient forecasting of hydrogen distributions for upcoming radio astronomy experiments including the Square Kilometre Array (SKA).	
Canadian Institute for Theoretical Astrophysics Summer Undergraduate Research Fellowship <i>Undergraduate Researcher</i>	Toronto, Ontario, Canada May 2023 – December 2023
Estimated neutron star tidal deformability from LIGO gravitational wave data using high-dimensional MCMC parameter estimation, and contributed new equation-of-state models to the collaboration's analysis pipeline.	
McGill Space Institute Summer Undergraduate Research Award <i>Undergraduate Researcher</i>	Montreal, Quebec, Canada May 2022 – April 2023
Explored integration of statistical priors into the power spectrum estimator for the Hydrogen Epoch of Reionization Array (HERA) collaboration's cosmic dawn experiment data pipeline, improving signal inference from radio astronomy observations.	

FEATURED PERSONAL PROJECTS

mercury_strategies [Private Repository] (work in progress)

- Developed an event-driven engine for capturing alpha in weather-derivative prediction markets (Polymarket), managing a dockerized infrastructure on AWS with integrated risk management. Features a provider-agnostic collection and analysis layer, asynchronous watcher architecture for real-time signal generation, and automated execution with velocity rotation, local orderbook tracking, and risk-aware position sizing via the Kelly Criterion. Includes 4 months of manual prediction market trading to gain market knowledge, intuition, and experience.

quant_core (https://github.com/kyleyhw/quant_core)

- Developed an open-core algorithmic trading platform with a flexible, modular architecture for strategy development and automated execution via Interactive Brokers (IBKR). Features a comprehensive backtesting engine with high-efficiency data caching and integrated risk management.

agent_evolve (<https://github.com/kyleyhw/agent-evolve>)

- Created a multi-model, cooperative evolutionary code-optimisation agent skill. Supervisor, explorer, and reviewer roles can each be filled by any LLM through CLI, with candidates evolving on isolated git branches. Features interactive visualization and surfaces the winner as a pull request for human approval.

HONOURS, AWARDS AND SCHOLARSHIPS

• Dean's List Scholar in the Faculty of Arts & Science	2022, 2023, 2024
• Canadian Institute for Theoretical Astrophysics Summer Undergraduate Research Fellowship (CITA SURF)	9,500 CAD, 2023
• Birkenshaw Family Scholarship II	1,000 CAD, 2023
• McGill Space Institute (now Trottier Space Institute) Summer Undergraduate Research Award (MSI SURA)	7,000 CAD, 2022
• David and Louise Fraser Scholarship	2,500 CAD, 2022
• University of Toronto Scholar	1,500 CAD, 2022
• Birkenshaw Family Scholarship	1,000 CAD, 2022

SELECTED COURSES

• Gravitational Waves & Numerical Relativity (Cambridge, 2025)	• Relativity Theory I (Toronto, 2023)
• Astrostatistics (Cambridge, 2025)	• Physics Project (Toronto, 2023)
• Cosmology (Cambridge, 2024)	• Computational Physics (Toronto, 2023)
• General Relativity (Cambridge, 2024)	• Computational Astrophysics (Toronto, 2023)
• General Relativity (Toronto, 2024)	• Advanced Classical Mechanics (Toronto, 2023)
• Relativity Theory II (Toronto, 2024)	• Geometry of Curves and Surfaces (Toronto, 2023)
	• Partial Differential Equations (Toronto, 2023)

SKILLS

- **Technical:** Scientific computing, machine learning, statistical inference, simulation, time-series analysis, data visualization, and high-performance computing, primarily in Python within Linux environments.
- **Financial Experience:** Trading in prediction markets and Counter-Strike skins.
- **AI Tools:** Claude Code, Google Gemini, Gemini CLI, ChatGPT, and Google Antigravity for automated software engineering, academic and scientific research, and coding.

LANGUAGES

- English (fluent, language of education)
- Cantonese (fluent, mother tongue)
- Mandarin (formally learned 10 years)
- German (formally learned 10 years)

RECENT MEMBERSHIPS

- Sidney Sussex College Football Team, Premier Division
- Institute of Astronomy Football Club
- Cambridge University Mountaineering Club
- Cambridge University Table Tennis Club

PERSONAL INTERESTS

- Rock climbing (9+ years experience both indoors and outdoors, with awards won at amateur, inter-high school and inter-university competitions. Also led a rock climbing extra-curricular activity during high school, including captaining the competition team)
- Soccer (7+ years league participation)
- Scuba diving (PADI open water certified)
- Travelling